

MATHEMATICS

TARGET : JEE- Advanced

CAPS-22

STRAIGHT LINE

SCQ (Single Correct Type) :

- Let $A(4, -1)$, B and C be the vertices of a triangle. Let the internal angular bisectors of angles B and C be $x - 1 = 0$ and $x - y - 1 = 0$ respectively. Let D , E and F be the points of contact of the sides BC , CA and AB respectively with the incircle of triangle ABC . The slope of BC is _____.
 (A) $\frac{1}{2}$ (B) 2 (C) 3 (D) 12
- Let $A(4, -1)$, B and C be the vertices of a triangle. Let the internal angular bisectors of angles B and C be $x - 1 = 0$ and $x - y - 1 = 0$ respectively. Let D , E and F be the points of contact of the sides BC , CA and AB respectively with the incircle of triangle ABC . If D' , E' , F' are the images of D , E , F in the internal angular bisectors of angles A , B , C respectively, then the equation of the circumcircle of $\triangle D'E'F'$ is _____.
 (A) $(x - 1)^2 + y^2 = 5$ (B) $x^2 + (y - 1)^2 = 25$
 (C) $(x - 1)^2 + (y - 1)^2 = 5$ (D) $x^2 + y^2 = 25$
- ABC is a triangle right angled at A with vertices A, B, C in the anti-clockwise sense in that order. $A = (1, 2)$, $B = (-3, 1)$ and vertex C lies on the X - axis. $BCEF$ is a square with vertices B, C, E, F in the clockwise sense in that order. ACD is an equilateral triangle with vertices A, C, D in the anti-clockwise sense in that order. The abscissa of centroid of $\triangle BCE$ is
 (A) -1 (B) $-\frac{1}{2}$ (C) $-\frac{1}{3}$ (D) $-\frac{2}{3}$
- Statement 1: Consider the point $A(0, 1)$ and $B(2, 0)$ and ' P ' be a point on the line $4x + 3y + 9 = 0$, then coordinates of ' P ' such that $|PA - PB|$ is maximum is $\left(-\frac{12}{5}, \frac{17}{5}\right)$
 Statement 2: $|PA - PB| \leq |AB|$
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct Explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
- The vertices of a triangle are $(1, \sqrt{3})$, $(2\cos\theta, 2\sin\theta)$ and $(2\sin\theta, -2\cos\theta)$ where $\theta \in \mathbb{R}$. The locus of orthocentre of the triangle is :



- (A) $(x-1)^2 + (y-\sqrt{3})^2 = 4$ (B) $(x-2)^2 + (y-\sqrt{3})^2 = 4$
- (C) $(x-1)^2 + (y-\sqrt{3})^2 = 8$ (D) $(x-2)^2 + (y-\sqrt{3})^2 = 8$
6. The true set of real values of a such that the point $M(a, \sin a)$ lies inside the triangle formed by the lines $x - 2y + 2 = 0$, $x + y = 0$ and $x - y - \pi = 0$, is
- (A) $(0, \pi)$ (B) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (C) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (D) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{2\pi}{3}, 2\pi\right)$
7. The equations $3x + 2y + 1 = 0$, $2x + 4y - 1 = 0$ and $3x^2 + 4xy + 4y^2 + 2x - 2y + 1 + \alpha = 0$ will have a unique solution if α equals
- (A) $\frac{2}{3}$ (B) $\frac{4}{5}$ (C) $\frac{3}{8}$ (D) $\frac{3}{4}$
8. If the points $(-2, 0)$, $\left(-1, \frac{1}{\sqrt{3}}\right)$ and $(\cos \theta, \sin \theta)$ are collinear, then the number of values of θ when $0 \leq \theta \leq \frac{\pi}{2}$.
- (A) 0 (B) 1 (C) 2 (D) Infinite
- MCQ (One or more than one correct) :**
9. Consider the three linear equations, $ax + by + c = 0$, $bx + cy + a = 0$, $cx + ay + b = 0$, where $a, b, c \in \mathbb{R}$. Which of the following is (are) correct?
- (A) If $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$, then the lines represent the entire XY plane.
- (B) If $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$, then the lines are concurrent.
- (C) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$, then the lines are coincident.
- (D) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$, then the lines are neither coincident nor concurrent.
10. The triangle formed by the lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ is
- (A) isosceles (B) scalene (C) acute angled (D) obtuse angled
11. Consider the straight lines $L_1 : x + y = 2$, $L_2 : 2x - y + 3 = 0$ and a variable point $P(a, a^2)$ where $a \in \mathbb{R}$. 'P' lies in the acute angle not containing the origin if 'a' lies in the interval
- (A) $(-4, -3)$ (B) $\left(\frac{3}{2}, \frac{5}{2}\right)$ (C) $\left(-2, -\frac{3}{2}\right)$ (D) $(5, 7)$
12. Consider the equation $y - y_1 = m(x - x_1)$. If m and x_1 are fixed and different lines are drawn for different values of y_1 , then :
- (A) the lines will pass through a fixed point (B) there will be a set of parallel lines
- (C) all the lines intersect the line $x = x_1$ (D) all the lines will be parallel to the line $y = x_1$.

13. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines $ax + by + c = 0$ are concurrent at :

- (A) $(1/2, 3/2)$ (B) $(-1/2, -3/2)$ (C) $(-1/2, 3/2)$ (D) $(1/2, -3/2)$

Comprehension Type Question:

Comprehension # 1

Paragraph for question nos. 14 to 16

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that $AE = AF$. Let P be a point inside the square ABCD.

14. The maximum possible area of quadrilateral CDFE is

- (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

15. The value of $(PA)^2 - (PB)^2 + (PC)^2 - (PD)^2$ is equal to

- (A) 3 (B) 2 (C) 1 (D) 0

16. Let a line passing through point A divides the square ABCD into two parts so that area of one portion is double the other, then the length of portion of line inside the square is

- (A) $\frac{\sqrt{10}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{11}}{3}$ (D) $\frac{2}{\sqrt{3}}$

Numerical based Questions :

17. The equation $9x^3 + 9x^2y - 45x^2 = 4y^3 + 4xy^2 - 20y^2$ represents 3 straight lines, two of which pass through the origin. Find the area of the triangle formed by these lines (in sq. units).

18. If the points $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right)$, $\left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right)$ and $\left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$ are collinear for three distinct values a, b, c and $a \neq 1$, $b \neq 1$ and $c \neq 1$, then find the value of $abc - (ab + bc + ac) + 3(a + b + c)$.

19. If the straight lines joining the origin and the points of intersection of the curve $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the x-axis, then find the value of $|k|$.

20. Is there a real value of λ for which the image of the point $(\lambda, \lambda - 1)$ by the line mirror $3x + y = 6\lambda$ is the point $(\lambda^2 + 1, \lambda)$? If so, find λ .

Matrix Match Type :

21. Let ABC be a triangle such that the coordinates of A are $(-3, 1)$. Equation of the median through B is $2x + y - 3 = 0$ and equation of the angular bisector of C is $7x - 4y - 1 = 0$. Then match the entries of column-I with their corresponding correct entries of column-II.

Column-I

- (A) Equation of the line AB is
(B) Equation of the line BC is
(C) Equation of CA is

Column-II

- (P) $2x + y - 3 = 0$
(Q) $2x - 3y + 9 = 0$
(R) $4x + 7y + 5 = 0$
(S) $18x - y - 49 = 0$

SUBJECTIVE:

22. Two vertices of a triangle are $(4, -3)$ and $(-2, 5)$. If the orthocentre of the triangle is at $(1, 2)$, then find the third vertex.
23. A pair of perpendicular straight lines is drawn through the origin and forming with the line $2x + 3y = 6$ an isosceles Δ right angled at the origin. Find the equation of the pair of straight lines and area of the Δ .
24. A straight line passing through $O(0, 0)$ cuts the lines $x = \alpha$, $y = \beta$ and $x + y = 8$ at A , B and C respectively such that $OA \cdot OB \cdot OC = 48\sqrt{2}$ and $f(\alpha, \beta) \leq 0$ where $f(x, y) = \left| \frac{y}{x} - \frac{3}{2} \right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$.
- (i) Find the point of intersection of lines $x = \alpha$ and $y = \beta$.
- (ii) Find the value of $(OA + OB + OC)$.
- (iii) Find the equation of line OA .
25. The vertices of a triangle OBC are $O(0,0)$, $B(-3,-1)$ and $C(-1,-3)$. Find the equation of line parallel to BC and intersecting the sides OB and OC , whose perpendicular distance from the point $(0,0)$ is $\frac{1}{2}$.

ANSWER KEY OF CAPS-22

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|---|----------------|---|-----------|----------|
| 1. (B) | 2. (A) | 3. (C) | 4. (D) | 5. (C) |
| 6. (C) | 7. (A) | 8. (ABC) | 9. (ABCD) | 10. (AD) |
| 11. (AD) | 12. (BC) | 13. (CD) | 14. (C) | 15. (D) |
| 16. (B) | 17. (30) | 18. (0) | 19. (1) | 20. (2) |
| 21. (A) | 22. $(33, 26)$ | 23. $5x^2 - 24xy - 5y^2 = 0$, $\frac{36}{13}$ sq. unit | | |
| 24. (i) $\alpha = 2$ and $\beta = 3$; (ii) $9\sqrt{2}$; (iii) $x - y = 0$ | | 25. $x + y + \frac{1}{\sqrt{2}} = 0$. | | |